

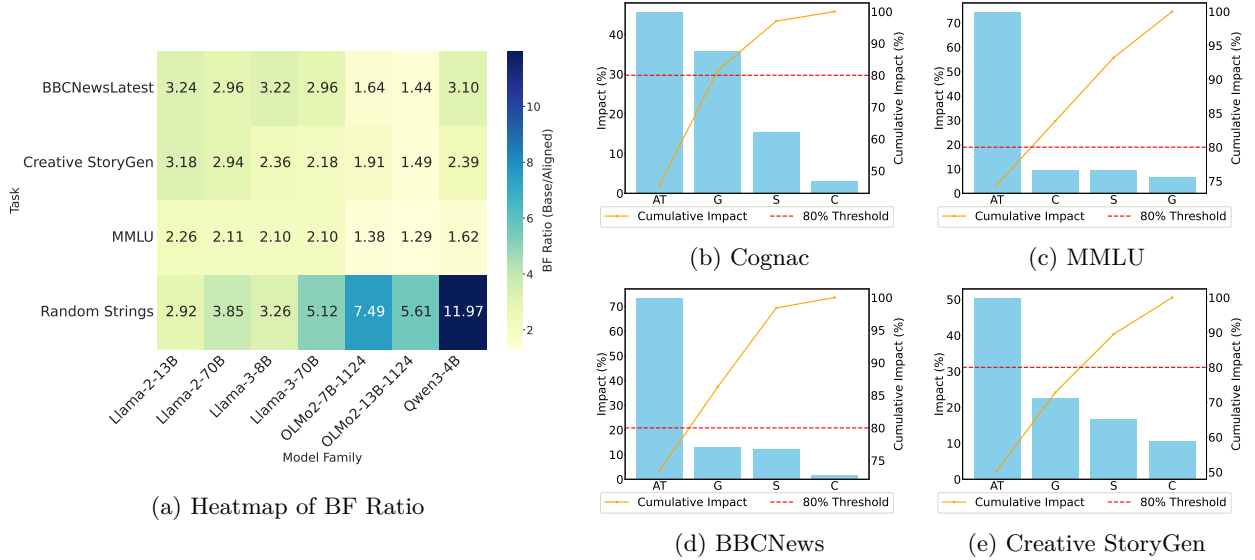


Figure 4: **Attributing BF Reduction.** (a) Heatmap of average BF ratio (Base/Aligned) across tasks and models. The numbers indicate the average ratio over all constraint levels. Note that for OLMo-2, we follow the convention in (OLMo et al., 2024) and treat the DPO version as the aligned model. (b)-(e) Pareto Analysis of BF across various IFs. *AT* indicates whether the model is aligned. *C* denotes the prompt complexity. *S* refers to model size, and *G* refers to model generation.

4.2 Pareto Analysis of BF

How dominant is this alignment effect compared to other factors? Figure 4a offers a first look: the Base/Aligned BF ratio is consistently high ($\gg 1$) across diverse tasks and models, peaking at 10 $\times$ for RANDOM STRINGS (task-wide) and appearing overall mildest for OLMo-2 (model-wide). To rigorously rank alignment against model size (*S*), generation (*G*), and prompt complexity (*C*), we perform a Pareto analysis (Figure 4) for Llama models. For each factor D_i , we define the unnormalized *Impact* $\tilde{I}(D_i)$ as the average absolute pairwise difference in BF when varying D_i while holding other dimensions constant:

$$\tilde{I}(D_i) = \frac{\sum_{d_i, d_j \in \text{Domain}(D_i), d_i \neq d_j} |\text{Avg}(\text{B}(\cdot | D_i = d_i)) - \text{Avg}(\text{B}(\cdot | D_i = d_j))|}{|\text{Domain}(D_i)| \times |\text{Domain}(D_i) - 1|}. \quad (9)$$

Then we normalize it as $I(D_i) = \frac{\tilde{I}(D_i)}{\sum_k \tilde{I}(D_k)}$.

The results crisply validate our intuition: **alignment tuning is the primary driver of BF reduction**. Across all tasks, it consistently crosses or approaches the 80% cumulative impact threshold, surpassing all other factors by a large margin.

Among the secondary factors, for tasks with richer inputs—such as MMLU (with more in-context examples) and BBCLATESTNEWS (with more headlines)—prompt complexity *C* and model size *S* emerge as the next most impactful. Prompt complexity *C* has a noteworthy effect: contrary to intuition, more context provided in the prompt does not always reduce BF but can in fact increase it, potentially due to the cognitive burden of processing complex linguistic structures. A detailed case study and comprehensive task-wise BF results are presented in §§ E and F. In contrast, for open-ended tasks like Cognac and Story Generation, model generation *G* plays a more dominant role, particularly improvements from Llama-2 to Llama-3. This shift likely reflects gains from the use of larger, more diverse datasets in training (Dubey et al., 2024).

5 Low Branching Factor Explains Generative Stability and Commitment

Our analysis in § 4 established that BF declines over the generation process (§ 4.1) and is significantly lower in aligned models (§ 4.2). This structural concentration of probability mass provides a unified

probabilistic explanation for three distinct generative behaviors: the insensitivity of aligned models to decoding hyperparameters, their reduced variance in majority voting, and the high performance cost of late-stage exploration.

5.1 Insensitivity to Decoding Configurations

Model developers adopt different decoding strategies when reporting LLM capabilities (Touvron et al., 2023; Dubey et al., 2024; Yang et al., 2024; Guo et al., 2025). The effectiveness of strategies like nucleus sampling (p) and temperature sampling (T) relies on the assumption that a diverse set of plausible next tokens exists. However, our BF analysis suggests that for aligned models, the "plausible set" is much smaller. If this holds, we expect aligned models to be robust to decoding choices, as there are few alternatives for the decoding algorithm to select even at higher temperatures.

We verify this by benchmarking decoding methods on MMLU-STEM (Hendrycks et al., 2021), extending prior work (Song et al., 2024; Renze & Guven, 2024; Shi et al., 2024a) to the latest models including DeepSeek-distilled models (Guo et al., 2025), which would generate long CoT before the final answer.⁸ Specifically, we evaluate model performance on MMLU-STEM (Hendrycks et al., 2021) under CoT prompting across different temperatures ($T = 0.6/1.0$) in temperature sampling and truncation thresholds ($p = 0.9/1.0$) in nucleus sampling (Holtzman et al., 2020). Further implementation details can be found in § A.

As shown in Table 1, aligned models exhibit limited performance variation (typically $< 10\%$) even when shifting from greedy-like settings to high temperatures. In contrast, Base models, which maintain higher BF, show significant sensitivity (up to 31%) to decoding parameters. Notably, DeepSeek-distilled Llama-8B, which generates long CoT and thus maintains a consistently low BF throughout generation, exhibits the smallest relative performance changes among 8B models. This empirically supports that *low BF effectively nullifies the impact of sampling method choices*.

Models	Default ($T=0.6, p=0.9$)	$T=0.6, p=1.0$	$T=1.0, p=0.9$	Min ($T=1.0, p=1.0$)	$\frac{\text{Default}-\text{Min}}{\text{Default}} \%$
Llama-3-70B-Instruct	78.50 (± 2.09)	77.60 (± 2.23)	77.50 (± 2.60)	75.90 (± 2.85)	3.31
Llama-3-70B	78.00 (± 3.52)	74.00 (± 3.80)	72.00 (± 4.38)	63.50 (± 5.02)	18.59
DeepSeek-R1-Distill-Llama-8B	66.30 (± 3.51)	65.70 (± 3.84)	62.70 (± 4.14)	59.70 (± 4.65)	9.95
Llama-3.1-8B-Instruct	63.00 (± 4.01)	61.50 (± 4.37)	57.50 (± 4.92)	50.50 (± 5.34)	19.84
Llama-3.1-8B	54.00 (± 4.61)	53.50 (± 4.92)	47.00 (± 5.21)	37.00 (± 5.48)	31.48

Table 1: **Experiment Results across decoding methods on STEM subset of MMLU.** We follow the common practice of using 5-shot CoT prompting. $\frac{\text{Default}-\text{Min}}{\text{Default}} \%$ indicates the maximum relative performance drop when deviating from the default decoding configuration.

Model	Maj@1 Std	Maj@3 Std	Maj@8 Std	Maj@16 Std	BF
DeepSeek-R1-Distill-Llama-70B	14.34	8.29	4.99	3.21	1.23
Llama-3-70B-Instruct	16.37	11.40	7.50	5.12	1.28
Llama-3-70B	27.78	19.53	13.22	9.23	1.31
DeepSeek-R1-Distill-Llama-8B	27.10	20.91	13.93	9.14	1.23
Llama-3.1-8B-Instruct	31.54	24.64	17.30	12.90	1.31
Llama-3.1-8B	36.41	29.78	20.43	14.05	1.35

Table 2: **Majority Voting@K standard deviation on MMLU-STEM with 200 samples.** We compute the standard deviation over 100 bootstrapping trials, each using 64 samples per instance. We set $T = 0.6, p = 0.9$ to match standard benchmarking settings, differing from $T = 1.0, p = 0.9$ setup in § 4. Lower temperature concentrates probability mass on fewer tokens, reducing BF and complicating direct comparisons. However, bootstrapping (100 runs) reveals minimal variability (≈ 0.01), confirming that the BF differences reported here remain significant. Consequently, BF remains a strong predictor of standard deviation.

⁸For Llama-3 series models, in our prior study, we find there is only a minor performance difference between Llama-3 and Llama-3.x. We mainly use Llama-3 in this paper as it includes the most diverse model collection.

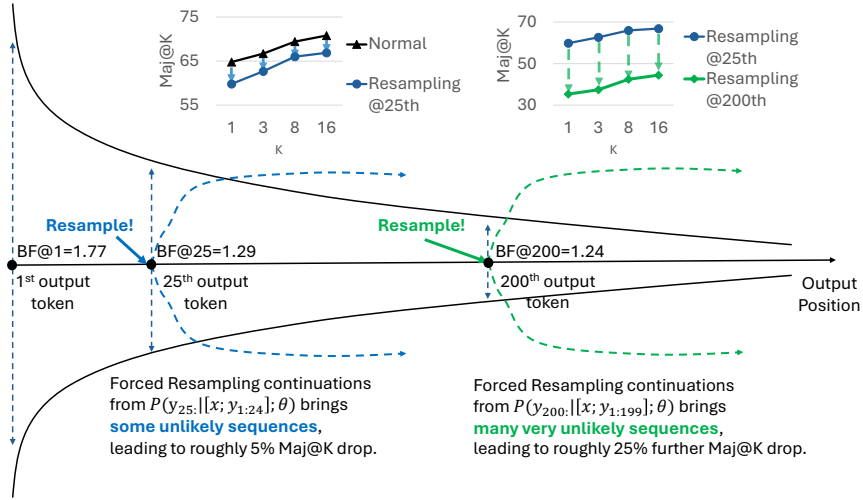


Figure 5: **Resampling from different output positions to assess the effect of interrupting BF reduction.** We resample new continuations at the 25th and 200th output token of DeepSeek-Distilled Llama-8B MMLU outputs. Results show substantial performance drops at both positions.

5.2 Reduced Variance in Majority Voting

If aligned models have a restricted output space (low BF), this should also manifest as reduced variance across independent samples. We test this hypothesis by evaluating output variance on MMLU-STEM using 200 samples per model. We benchmark the standard deviation across Majority@K accuracy metrics ($K = 1, 3, 8, 16$) with $T = 0.6, p = 0.9$.

As detailed in Table 2, BF serves as a strong predictor of sampling consistency. The Long-CoT models (DeepSeek-R1-Distill), which generate significantly longer outputs and achieve the lowest global BF, consistently show the smallest performance variance. This confirms that the “stability” observed in decoding benchmarks is not an artifact of specific hyperparameters, but a fundamental property of the narrowed generation manifold.

5.3 Risks of Mid-Generation Forking

The stability provided by low BF suggests a strong commitment to a specific reasoning trajectory. Does this commitment imply that the model cannot effectively explore alternative paths once generation has begun? If BF reflects a semantic “lock-in,” forcing the model to branch out (fork) at late, low-BF stages should disrupt coherence and degrade performance.

To examine this, we conduct a resampling experiment using DeepSeek-Distilled Llama-8B output samples. Procedurally, for a given position t : 1) Take the prefix $y_{<t}$ generated by the model. 2) Sample a new continuation $y'_{\geq t}$ from $P(Y_{\geq t} | [x, y_{<t}]; \theta)$. 3) Evaluate the full sequence $[y_{<t}, y'_{\geq t}]$ on the task.

As shown in Figure 5, performance drops sharply when resampling occurs at a later, lower-BF position in the sequence. This suggests that aligned models are not just concentrating probability mass locally (reflects a “deeper commitment” to specific paths), but are actively locking into trajectories, making late-stage deviations more error-prone. In practice, this highlights a key application of BF: *parallel sampling should be applied early, while BF remains high*, to ensure meaningful diversity and avoid quality degradation.

6 How does Alignment Tuning Impact BF?

Why does alignment tuning exert such a pronounced effect on BF? Building on the superficial alignment hypothesis (Zhou et al., 2024) (“*Alignment tuning might simply teach base LLMs to select a subdistribution of data formats for interacting with users.*”) and recent tuning-free alignment work (Lin et al., 2023; Fei et al., 2024; Lake et al., 2025), we hypothesize base models already encode low-entropy conditional distributions. In